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Superseding
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NAVORD
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MILITARY SPECIFICATION
CARTRIDGES, 20MM, FOR
AIRCRAFT GUNS MK 11 AND MK 12

1. Scope

1.1 This specification covers the assembly, testing,inspection and packing of cartridges for the MK 11 and MK 12 Aircraft Guns.

1.2 Classification. The cartridges covered by this specification are the following:

a. Target Practice (TP) MK 105 (LD 289041)
b. High Explosive Incendiary (HEI) MK 106 (LD 289042 and LD 289043)
c. Armor Piercing Incendiary (API) MK 107 (LD 289044)
d. Armor Piercing Tracer (APT) MK 108 (LD 289045)

2. Applicable Documents

2.1 Government Documents. The issue in effect of the following documents on date of invitation for bid or request for proposal, form a part of this specifications to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-250/4 - Aluminum Alloy 2024, Plate and Sheet

Military

MIL-A-625 - Ammunition, Small Arms, General Specification for

MIL-I-45607 - Inspection equipment, Supply and Maintenance, for Ordnance

MIL-P-16594 - Projectiles, Bombs, Rockets and Guided Missile Warheads, Inert Parts

MIL-T-19314 - Tracer, MK 20 Mod 0

MIL-P-1394 - Primer, Electric, M52A3B1

MIL-P-19973 - Projectiles, Minor Caliber

MIL-C-45662 - Calibration System Requirements

FSC 1375

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MIL-P-18903 - Projectiles, Armor Piercing, Minor Caliber

MIL-Q-9858 - Quality Program Requirements

STANDARDS

Bureau of Ordnance

O.S. 5944 - Fuze, MK 78

A.S. 1052 - Propellant for 20MM Naval Aircraft Guns

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-331 - Fuze and fuze components, environmental and performance tests for

MIL-STD-414 - Sampling Procedures and Tables for inspection by variables for percent defect

MIL-STD-1167 - Ammunition Data Card

MIL-STD-1314 - Safety Precautions for Explosive-Loaded Items

DRAWINGS

Navy Dept. - Bureau of Naval Weapons, Naval Air Systems Command

BUORD 1170771 - Fuze, MK 78, Requirement and Test Procedures

BUORD LD 284914 - Fuze, MK 78, Mod 1

BUORD LD 256092 - 20MM Ammunition (for Aircraft Gun MK 12) index of Loaded Assemblies, Drawings and Specifications

BUORD LD 550023 - Fuze, MK 78, Mod 2

BUORD LD 289041 - Cartridge, 20MM, Target Practice (TP), A/C, MK 105, List of Component LD's, Drawings and Specifications

BUORD LD 289042 - Cartridge, 20MM, High Explosive Incendiary (HEI), MK 106 Mod 0, List of Component LD's, Drawings and Specifications

BUORD LD 289043 - Cartridge, 20MM, High Explosive Incendiary (HEI), MK 106 Mod 1, List of Component LD's, Drawings and Specifications

BUORD LD 289044 - Cartridge, 20MM, Armor Piercing Incendiary (API), A/C MK 107, List of Component LD's, Drawings and Specifications

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BUORD LD 289045 - Cartridges, 20MM, Armor Piercing Tracer (APT), A/C, MK 108 List of Component LD's, Drawings and Specifications

P.S. 7548066 - Primer, Electric, M52A3B1 Assembly

BUORD LD 266027 - 20MM Gun Mechanism, MK 12, List of Drawings, Sub-Assemblies, etc.

BUORD LD 566057 - 20MM Gun Mechanism, MK 11, List of Drawings, Sub-Assemblies, etc.

Army ORD Dwg. D7553247 - Mann Barrel, Action, Test, Electric, Caliber .60 and 20MM

BUORD 564225 - 20MM Ammunition Packaging, Method of Marking

BUORD LD 256008 - Ammunition Packaging (for 20MM Gun MK 12), List of Drawings, Sub-Assemblies and Parts

Dept. of Army - Ordnance Corps.

49-6-48 - Rest. Recoil, Multiple Purpose, Caliber .60 and 20MM

D7553269 - Action, Test, Caliber .60 and 20MM

FD-16042 - Primer Firing Device, Electric, for Ammunition Testing

SK-232629 - Test Barrel (Caliber .60 and 20MM) for T130 20MM Round

PUBLICATIONS

Navy Dept. - Bureau of Naval Weapons

OP-400 - General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment

OP-2574 - 20MM Barrel Erosion Gage Kit M10 (T23)

OP 2719 Vol. 1 - 20MM Aircraft Gun Pod, MK 4 Mod 0 Description, Operation and Maintenance

Dept. of the Army - Ordnance Corps.

ORD-M608-PM - Ordnance Proof Manual (Vol. III)

AMCR 715-505-Vol. 8 Oct. 1965 - U.S. Army Material Command, Ammunition Ballistic Acceptance Test Methods, Test Procedures for 20MM Cartridges

(Copies of specifications, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENT

3.1 GENERAL

3.1.1 Item Requirements. The individual item requirements shall be as specified herein, and as stated in the Detail Specification for the applicable individual

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component; failure to comply with the requirements of either will result in rejection of the item.

3.1.2 Before assembly of the cartridge, the supplier shall insure that all components are clean, free from grease, dirt, rust, chips and other foreign material. When cleaning is required the method employed shall be satisfactory to the Government. No method or solvent shall be used which would injure the parts in any way.

3.1.3 Components and materials furnished by the government shall be given a careful visual inspection by the supplier before assembling, to prevent the possible use of any damaged or obviously defective parts (dents, scratches, etc.). If any suspect parts are found, they shall be submitted to the Government for examination and decision as to their fitness for loading and assembling, and their compliance with applicable specifications and drawings.

3.1.4 Method. The method of projectile loading and cartridge assembly shall be uniform for each lot and in accordance with applicable drawings. Once a satisfactory production rate has been established, no significant change in projectile loading or cartridge assembly processes shall be made without notifying the cognizant field activity and the procuring agency.

3.1.5 Test Equipment and Gages. All test equipment and gages shall be supplied and maintained by the supplier as specified in Specification MIL-I-45607. The supplier shall maintain a calibration system as specified in Specification MIL-C-45662, unless otherwise specified in the contract. However, a supplier currently under the requirements of MIL-Q-9858 shall be governed by those requirements for items procured in accordance with this specification.

3.2 Propellant Charge. The quantity of propellant making up the charge for each cartridge may be measured volumetrically. Production weighting shall be suitably controlled using average (X) and range (R) charts to insure that actual weight is within plus or minus five grains of the assigned weight for the index of powder being used. Any deviations from the exact charge weight shall be randomly distributed over the range to minimize the effect of this variation. Great care shall be taken to load all charges as close to the exact charge weight as possible.

3.2.1 Propellant Temperature at Time of Filling. Propellant shall be conditioned at room temperature prior to opening of tanks. Propellant tanks shall not be opened for cartridge filling when the room temperature is within five degrees of the dew point.

3.3 Primer

3.3.1 Primer Insertion. The primers shall be inserted in the cartridge cases as shown on the applicable drawings. Graphite or lubricants shall not be used to facilitate primer insertion. Care should be exercised during the primer insertion and crimping operation to insure that the primer is not canted, inverted or damaged.

3.3.2 Electrical Resistance of Primed Cartridge Case. Each primed case shall be tested for electrical resistance using an approved resistance measuring circuit after crimping and prior to primer sealing and assembly into cartridges. The electrical resistance of each primed cartridge case shall be in accordance with the applicable specification for the primer. Tests must be as specified in 4.4.1.

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3.3.3 Primer Sealing. In waterproofing the joint between the contacting surfaces of the primer and cartridge case, an adequate amount of purple lacquer of suitable viscosity is applied over the entire interface between the cartridge case and the primer wall in order to effect a waterproof seal. No purple lacquer shall be applied to a 0.140 inch diameter at the center of the primer button. The lacquer shall have sufficient purple color to permit ready visual identification of its presence around the joint.

3.4 Projectile Insertion and Crimping. During assembly of the projectile into the cartridge case, the joint between case and projectile shall be waterproofed and crimped in accordance with applicable drawings. The crimping process shall be controlled to insure that the maximum pressure applied to the neck of the cartridge case by the crimping medium shall not exceed 39,000 p.s.i. The assembled cartridge shall meet the moisture-proof and extraction force requirements of 3.7.2, 3.7.4 and 3.5.

3.5 Projectile Extraction. The projectile extraction test, as specified in 4.4.2 shall be conducted on a sample randomly selected from each lot. The force required to extract the projectile must be 1300 pounds minimum with process control by (X) and (R) chart recordings.

3.6 Cartridge Gaging. The supplier shall gage each cartridge in the profile and alignment gage (chamber gage) Drawing No. 1314850 and height gage for the cartridges as specified by the procuring agency. If a cartridge fails to enter the profile and alignment gage freely or if its overall length is not within specified tolerance, it shall be rejected, subject to correction and retest.

3.7 Ballistic Test. The cartridge shall meet the following requirements in the ballistic test.

3.7.1 Velocity. The corrected average muzzle velocity as determined by test specified in 4.4.3.1 shall be within plus or minus (±) 50 feet per second of the velocity for the assigned charge weight for the index of propellant in the cartridges and the standard deviation in the reading shall not exceed 40 feed per second.

3.7.2 Waterproof Velocity. The corrected average muzzle velocity when tested in accordance with 4.4.3.2 shall be within plus or minus (±) 75 feet per second of the velocity for the assigned charge weight for the index of propellant in the cartridges and the standard deviation in the readings shall not exceed 40 feet per second.

3.7.3 Accuracy. The average mil value (M) of samples fired as specified in 4.4.4.5 shall not exceed the values indicated:

a. HEI Cartridge, MK 106	0.95 mils
b. API Cartridge, MK 107	0.95 mils
c. APT Cartridge, MK 108	1.20 mils
d. TP Cartridge, MK 105	1.39 mils

3.7.4 Action Time. The average time spread (R) as determined in 4.4.3.3 shall not exceed 0.33 millisecond. Individual cartridges must exhibit an action time less than or equal to 3.3 milliseconds.

3.7.5 Tracer Functioning. The tracer must function with a minimum burning time of 1.1 seconds and without the separation of the tracer cup from the pro-

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jectile or tracer incendiary composition from the tracer cup when tested as specified in 4.4.3.6. Separation of the tracer including composition or Sealant from the cup shall be determined by means of a .032 inch thick aluminim plate (2024-T) 5' x 5' located 15' from the muzzle. Penetration of the plate by either the sealant, the entire incendiary charge, or by any fragments thereof will be cause for rejection. Separation of one tracer cup will reject the lot.

3.7.6 Fuze Sensitivity. The sensitivity of the fuze must be such that the projectile produces a high order burst by plate impact when tested as specified in 4.4.3.7.

3.7.7 Incendiary Ignition. The incendiary ignition must be such that a projectile impacted on a target, when tested as specified in 4.4.3.8 produces ignition of the kerosene. The term ignition is defined as rendering the kerosene or fumes thereof luminous.

3.7.8 Pressure Firing. The average corrected peak pressure shall not exceed 58,000 p.s.i. (copper) and the standard deviation shall not exceed 3,000 p.s.i. (copper). All results are to be recorded.

3.7.9 Function and Casualty. The cartridge shall function in the service weapons properly and at a rate of fire of not less than 925 rounds per minute (MK 12 Gun) and 3800 rounds per minute (MK 11 Gun - MK 4 Pod) when tested as specified in 4.4.3.9. Function and casualty firing defects plus all other firing defects except fuze sensitivity (3.7.6), incendiary ignition (3.7.7), and tracer functioning (3.7.5) shall not exceed the limits specified in Table I.

TABLE I

FIRING DEFECTS AND NO. OF DEFECTIVE UNITS PERMITTED IN A LOT

Defects	Maximum No. of Defective Units Permitted (1)
Misfire	0
Projectile remaining in bore	0
Premature bust (2)	0
Premature ignition (3)	0
Primer leak:	
a. Large leak, or evidence of gas erosion on base of case or bolt face of gun.	2
b. Small leak, or evidence of gas leakage around primer body.	5

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c. Damaged or blown primer (4)	0
Case casualty:	
a. Circumferential rupture	
(1) Complete at crimping groove	0
(2) Partial; neck, shoulder, body or base	0
b. Longitudinal split	
(1) Shoulder or body	0
(2) Neck	3
c. Flange cracks (6)	
Failure to extract, including sheared rim (5)	0
Failure to chamber	0
Gun stoppage (from ammunition defects not listed)	0
Separation of metal parts of projectile assembly in Flight	0

NOTE:

(1) The number of defective units permitted pertain to the total number of rounds fired as prescribed in Table I, with the exception that primer leaks shall not be considered defects for rounds fired in a single action.

(2) Premature burst is defined as functioning of an HEI projectile in the gun or in flight before impact on target.

(3) Premature ignition is defined as functioning of an API projectile in the gun or in flight before impact on target.

(4) "Unseated" or "upset" primer buttons with no indications of "gas leakage" are not to be considered defects that will reject the lot, and reporting of such defects on Acceptance Test Reports is not required.

(5) Prior to firing, a light application of oil or other lubricants as required may be applied to the cartridge case (teflon coated case excluded).

(6) Flange cracks, as determined by visual inspection, shall not be cause for rejection of the lot but shall be recorded and reported.

3.8 Safety. There shall be no bore or flight premature or tracer cup separations when tested as specified in 4.4.3.10.

3.9 Drop. There shall be no flashing or indication of ignition of the incendiary load from the MK 107, Armor Piercing Incendiary, cartridges and there shall be no indication of detonation or deflagration of either the fuze or the projectile of the MK 106, High Explosive Incendiary cartridges when tested as specified in 4.4.3.11.

3.10 Workmanship. The special features of these cartridges which establish their service value, are obtainable only by exercising care in each and every detail of loading and assembly. All operations must be carried out in accord-

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ance with the applicable drawings and specifications.

4. Quality Assurance Provisions and Test Requirements.

4.1 "Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections set forth in this document where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements".

4.1.1 Lot Sizes

4.1.1.1 Pilot Lot. Each new supplier shall furnish 5000 cartridges of a type as a pilot lot. This pilot lot shall be assembled in the plant in which the supplier is to fill his contract, and shall be assembled under the same process that he will follow in executing the contract. This lot shall be forwarded as directed by the procuring agency for test in any manner considered necessary. The results of this test will determine whether or not the supplier is ready to proceed with his contract. Failure of this pilot lot to give satisfactory test results may be considered sufficient cause for cancellation of the contract. The cartridges referred to in this paragraph shall be provided at the supplier's expense.

4.1.1.2 Production Lot. The size of production lots shall not be less than 20,000 rounds, and not exceed 50,000 rounds. One lot shall contain:

a. Projectiles of one lot
b. Cartridge cases of one lot
c. Primers of one lot
d. Propellant powder of one index
e. Assembled fuzes of one lot
f. Tracers of one lot

The component lot having the smallest number after rejects are removed shall control the actual lot size. Ammunition data cards, Figure I, (MIL-STD-1167) shall be printed for each lot.

4.1.1.3 Mixed Component Lot. When the lot size has been established, all components that are left over are to be retained until a sufficient number is accumulated for a mixed lot of 20,000 to 50,000 cartridges. The accumulated leftovers of propellant shall not be used in mixed component lots unless blended by a method acceptable by the procuring agency and a change determination shall be made for its assembly into cartridges. The ammunition data card (Figure I) shall contain information on all lots of components used.

4.1.1.4 Inspection Lot. The supplier shall comply with the random sampling methods with Standard MIL-STD-105.

4.2 Sampling

4.2.1 Size of Ballistic Samples. The size of the sample of the various ballistic acceptance tests shall be as shown in Table II. The ballistic test samples shall be selected randomly by the Government and forwarded to the test activity, designated by the procuring agency for ballistic tests.

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TABLE II

BALLISTIC TEST SAMPLES

Test	Cartridge Type	Amount	Requirements
Velocity	(All)[4]	25	3320 ± 50 ft/sec
Water Proof Velocity[1]	(All)	20	3320 ± 75 ft/sec
Accuracy	(All)	20	HEI, API .95 mils; APT, 1.20 mils, TP 1.39 mils
Action Time	(All)	20	R .33 milliseconds Indiv. R 3.3
Tracer Function	(APT)	20	Tracer cup separation: Accept on 0 reject on 1 Dud (burning time less 1.1 sec): Accept on 3 reject on 4
Fuze Sensitivity	(HEI)	20	Accept 1 reject 2
Incendiary Ignition	(API)	20	Accept 1 reject 2
Function & Casualty[2]	(All)	300	Table I
Safety[3] (HEI, API, APT)		20	Accept 0 reject 1
Drop	(HEI, API)	10	Accept 0 reject 1

[1]In the event of a misfire an additional round shall be fired in lieu of the failing unit, and the number of rounds fired for functioning and casualty test accordingly.

[2]Will be equally divided for firing in both MK 11 and MK 12 Guns.

[3]Assembled projectiles only, to be furnished in this sample. Additional components will be supplied by the Test Activity.

[4]"All" means that all cartridge types covered by this specification will be subjected to this test. The designation of a particular type (API, HEI, etc.) means that the test is for that type of cartridge only.

4.3 Inspection

4.3.1 Place. Unless otherwise specified, the sampling inspection, and tests of ammunition shall be made at the place of manufacture by personnel employed and furnished by the Government. The supplier shall provide adequate facilities, such as necessary floor space, tables and benches.

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AMMUNITION DATA CARD — FIXED AMMUNITION

Caliber: 20 MM
Type:
For Guns:
Complete Round Drawing No:
Powder Index: SPDN
Muzzle Velocity: (Automatic 3290 to 3350 fps) Pressure (Cu.) 58,000 psi
Assembled by: N. A. D., McAlester
P/O Order No:
Ctg. Mk. Mod.

Amm. Lot No:
Quantity in Lot:
Charge Weight:
Date:
Job Order No:
Packed: in a box

Description of Components	Mark and Mod. or Type	Manufacturer	Lot Number	Year Mfd.	Date Loaded	Drawing Number
Primer						
Cartridge Case						
Propellant Powder						
Projectile						

REMARKS:

Figure 1

4.3.2 Plant Access. The Government shall be allowed access to those parts of the plants where work for the Government is being done at any time when such work is in progress and shall be allowed to observe and check any and all processing operations and preparations for shipment without interfering with production.

4.3.3 Component Assurance. Before the components are assembled into cartridges and after such components have been withdrawn from store and stock on hand, the Government shall, without interfering with production, assure in such manner as appears practicable that the components are suitable in all respects for assembly into satisfactory cartridges.

4.4 Tests. Compliance with the requirements specified herein is mandatory prior to acceptance, by the Government, of lots represented by the samples submitted for testing. Failure to comply will result in rejection of the lot.

4.4.1 Electrical Resistance Test. The electrical resistance of each primed cartridge case, measured with a circuit approved by the Government shall not be less than 1000 nor more than 5,000,000 ohms. Instrumentation shall be of the Multimeter AN/PSM-4 type or equivalent, designed to pass a maximum current of 25 milliamperes when the test leads are shorted.

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4.4.2 Projectile Extraction Test. Each cartridge from the test sample shall be tested in a tensile test machine or equivalent machine of suitable capacity which will register accurately the force required to extract the projectile from the cartridge case, after it has been crimped, when the rate of travel of the test head is controlled within 0.125 to 0.25 inch per minute, inclusive. Lot acceptance or rejection in respect to requirement of 3.5 shall be judged in accordance with the double sampling plan outlined below (AQL 0.30%):

Sample	Sample Size	Cumulative Sample Size	Accept Number	Reject Number
1st	50	50	0	2
2nd	100	150	1	2

The results are to be recorded.

4.4.3 Ballistic Tests. The test activity selected by the procuring agency shall conduct the following tests to insure compliance with 3.7:

4.4.3.1 Velocity. Twenty-five cartridges of the test sample shall be selected at random from all of the ammunition boxes in which the samples were shipped and fired for velocity from the automatic test weapons specified in 4.7. If adequate instrumentation is not available at the test facility for measuring velocities when firing fully automatic, the test may be conducted in any test weapon specified in 4.7. The method of test shall be in accordance with specification MIL-A-625. Publication ORD-M608-PM, Vol. III and AMCR 715-505, Vol. 8. Ten reference or control rounds shall be assembled by the test activity from selected components and master propellant and interspersed with the last twenty test cartridges. The corrected average velocity, standard deviation and rate of fire of the last 30 rounds of the burst shall be computed for conformance with 3.7.1. The first five rounds are to be considered warming rounds. The average corrected velocity for the test cartridges shall be computed by the following formula:

$V=V_1+(V_2-V_3)$ when

V_1=Corrected muzzle velocity measured on 20 test cartridges.

V_2=Assigned muzzle velocity of the propellant used in reference cartridges.

V_3=Average muzzle velocity measured on 10 reference cartridges in the test weapons.

When the velocity test results do not meet the criteria, chamber pressure testing shall be performed in accordance with 4.4.3.4.

4.4.3.2 Waterproof Velocity. Twenty cartridges of the test sample shall be randomly selected and submerged in water for 24 hours in accordance with Publication ORD-M608-PM, Vol. III and AMCR-715-505, Vol. 8. The rounds are then to be fired from the same test weapon specified in 4.7. Ten reference rounds as described in 4.4.3.1 shall be fired from the same weapon and on the same day as the test cartridges. The corrected average velocity and standard deviation shall be computed for conformance with 3.7.2. The corrected average velocity will be computed from the formula in 4.4.3.1.

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4.4.3.3 Action Time. Twenty test sample cartridges shall be selected and submerged in water for 24 hours prior to firing. Action time shall be evaluated on the basis of single action firing of four subsets composed of five rounds each in a test weapon (see 4.7). Subsets shall consist of rounds in the order fired and a range value (R) shall be determined for each subset. The average round ($\bar{R}$) shall then be computed as follows:

$$\bar{R}=\frac{R_1+R_2+R_3+R_4}{4}$$

Where each of the R_1 through R_4 values represents the range of a subject of given readings. The test may be conducted in conjunction with 4.4.3.2. In the event the requirements of 3.7.3 are not met, another twenty round sample (not subjected to water) shall be fired for conformance with 3.7.3.

4.4.3.4 Pressure Firing. Each cartridge of the test sample shall be fired in a pressure barrel in accordance with publication ORD-M608-PM, Vol. III and AMCR 715-505, Vol. 8. At least 3 warming rounds, same as those tested, shall be fired. The average peak pressure shall be corrected for the condition of the barrel by the following formula:

$P=P_1+(P_r-P_b)$ when

P =Corrected peak pressure of the test cartridge.

P_1=Average peak pressures measured on test cartridge.

P_r=Assessed pressure of referenced cartridges.

P_b=Average pressure of 10 reference cartridges in the pressure barrel. This firing of the reference round sample shall be interspersed throughout testing.

This test to be performed only in the event of failure in velocity firing test or waterproof velocity firing tests, 4.4.3.1 and 4.4.3.2.

4.4.3.5 Accuracy Firing. Two groups of ten cartridges each shall be fired to determine the average disposition value from an accuracy barrel as specified in 4.7 at a vertical target placed at a distance of at least 500 yards. Three warming rounds should be shot if the gun is cold. The test rounds shall be fired at a uniform rate. For each group of ten cartridges fired, the vertical and horizontal patterns of the impacts on the target shall be determined. The average pattern ($\bar{P}$) shall be computed as follows:

$$\bar{P}=\frac{P_1+P_2+P_3+P_4}{4}$$ where

P_1=The longest vertical distance in inches between any two impacts of the first group of ten rounds fired.

P_2=The longest horizontal distance in inches between any two impacts of the first group of ten rounds fired.

P_3=The longest vertical distance in inches between any two impacts of the second group of ten rounds fired.

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P_4 = The longest horizontal distance in inches between any two impacts of the second group of ten rounds fired.

All measurements shall be made from center of impact holes. The average pattern (P) shall be converted to mils as follows:

$M = \frac{P}{d} \times 1000$ where

d = The distance in inches from the gun muzzle to the target. The average mil value (M) computed as specified should not exceed the average mil limit specified in 3.7.3 for each cartridge type. The average mil value (M) shall be reported for information purposes.

4.4.3.6 Tracer Functioning. Twenty cartridges from the test sample shall be selected and submerged in water 24 hours prior to firing. The cartridge shall be fired from any test weapon specified in 4.7. A suitable screen to detect thrown tracer cups shall be placed near the gun muzzle. A suitable material is 1/4" pressed paper board (4' x 4') placed at a distance of 15 to 30 feet from the gun muzzle. Gun elevation shall be such that complete burning of the tracer can be observed prior to impact of the projectile. Suitable aids such as binoculars and stop watches with sweep second hands should be used to insure that complete and accurate times are obtained. All burning times, the number of duds and tracer cup separations shall be recorded and computed as necessary to determine conformance with 3.7.5. Lot disposition for duds shall be determined in accordance with the AQL of 4.0% defective of MIL-STD-105, single sampling plan. Lot disposition for burning time less than 1.1 seconds shall be determined in accordance with an AQL of 6.5% defective of MIL-STD-105, single sampling plan. This test may be conducted in conjunction with 4.4.3.2.

4.4.3.7 Fuze Sensitivity. Twenty cartridges shall be selected from the test samples and submerged in water for 24 hours prior to firing. The cartridges shall be fired from any test weapon specified in 4.7 for conformance with 3.7.6. The projectile must impact against a target of 0.032 inch thick aluminum plate (AL-2024-T3, QQ-A-250/4) normal to the line of fire at a range of 150-300 feet. Each projectile shall impact at least three inches from previous impact perforations. Lot disposition shall be determined in accordance with an AQL of 4.0% defective of MIL-STD-105, single sampling plan. When feasible, this test may be conducted with 4.4.3.2.

4.4.3.8 Incendiary Ignition. Twenty cartridges shall be selected from the test samples and submerged in water for 24 hours prior to firing. The cartridges shall be fired from any test weapon as specified in 4.7 to impact against an aluminum alloy (AL-2024-T3) target 1/16" to 1/8" thick normal to the line of fire at a range of 125-300 feet for conformance with 3.7.6. A 12-inch square tray filled to a depth of 1/4 inch with kerosene shall be placed approximately 3 inches behind the target and parallel to and not more than 6 inches below the projected line of fire. Observations shall be made to determine whether or not a projectile impacted on the target produces ignition of the kerosene. The decision of the test director will be final in determining whether or not ignition occured on impact. Lot disposition shall be determined in accordance with an AQL of 4.0% defective of MIL-STD-105, single sampling plan. When feasible, this test may be conducted in conjunction with 4.4.3.2.

4.4.3.9 General Function and Casualty Test. Cartridges of the test sample

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shall be fired, fully automatic, in either or both automatic weapons specified in 4.7. Suitable targets must be located to detect any metal parts separation or break-up of the projectile assembly. In general the test shall be in accordance with sub-section 7-15 of Publication ORD-M608-PM, Vol. III and AMCR 715-505, Vol. 8. From the test samples of the pilot lot and all production lots submitted for ballistic acceptance, fifty (50) cartridges shall be temperature conditioned at +165°F and fifty (50) cartridges shall be temperature conditioned at -65°F for a minimum of two hours prior to firing. Temperature conditioning need apply only to firings in the MK 12 gun. A record of all discrepancies should be maintained throughout the testing to insure conformance with 3.7.9.

4.4.3.10 Safety. The projectiles furnished for this test shall be assembled into cartridges by the testing activity utilizing station components and a propellant charge that would produce a chamber pressure of 68,000 plus or minus (+) 2000 p.s.i. (copper) in a new gun at +70°F. The cartridges shall be fired from any test weapon as specified in 4.7. Also projectiles shall be observed to a point of impact at least 2500 yards from the gun for premature bursts, premature ignitions, and tracer cup separations shall be noted and reported.

4.4.3.11 Drop. Ten cartridges selected from the test samples shall be dropped nose down, onto a steel plate from a height of 40 feet for conformance with 3.9.

4.5 Component Failures. When any of the components furnished by the Government are found to be the cause of failure or malfunctioning of the cartridges, such cartridges shall be held for disposition by the procuring agency, who will direct such tests as are considered necessary to determine the final disposition of such cartridges.

4.6 Retests. The authorization for retest and/or additional testing, and the number of rounds to be tested, will depend on the circumstances of the individual case, and will be at the discretion of the procuring agency.

4.7 Test Weapons. The 20MM machine guns, MK 12 and MK 11, used in conducting ballistic tests requiring automatic fire shall be known to render satisfactory performance and in all respects shall be equivalent to the current service weapons. Sufficient check-out rounds should be fired to determine weapon performance. The rounds may be any type of the 100 series from a previously accepted ammunition lot or reference rounds. The guns selected for velocity firing test 4.4.3.1 shall be equipped with barrels which will give an average muzzle velocity of not more than 100 feet per second below that obtained from an average new barrel of the type being used. The muzzle velocity shall be determined by firing a minimum of 10 reference rounds. The test weapons shall conform to the following:

MK 11 Gun - Assembled in accordance with BUORD LD 522057

Cyclic rate of fire - 4000 plus or minus (+) 200 rounds per minute
Erosion limits (measured in accordance with OP-2719, Vol. 1):

Barrel - 0.025 inch maximum
Barrel Inserts - 0.025 inch maximum
Rifled Inserts - 0.010 inch maximum

MK 12 Gun - Assembled in accordance with BUORD LD 172525 (MOD 0) and LD 266027 (MOD 3)

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Cyclic rate of fire - 925 rounds per minute (minimum)

Lock clearance - 0.010 inch maximum
Breeching Space - 5.118 inch plus 0.007 minus 0.005

Barrel erosion limits (measured in accordance with OP 2574):

Barrel, MK 11, Mod 3 0.046 inch (maximum)
Barrel, MK 11, Mod 4 0.057 inch (maximum)

Mann Barrel - The Mann barrel counterpart of the MK 11 machine gun barrel assembled in accordance with Army Ordnance Dwg. D7553247 (Action Test, Electric, Caliber .60 and 20MM) shall be used in acceptance testing of [illegible] cartridges where specified. The barrel shall be disqualified for use when the muzzle velocity has dropped 100 feet per second below that of an average new barrel (3320 feet per second) when firing reference rounds.

4.8 Defective Weapons. When any firing defect is found to have been caused by the weapon used, such firing defect shall not be counted against the ammunition but shall be corrected or replaced and the test is repeated.

5. PREPARATION FOR DELIVERY

5.1 Reservation and Packaging. Not applicable

5.2 Packing. (See Section 6) - Cartridges shall be packed in accordance with one level of packing as follows:

5.2.1 Level A. Cartridges shall be packed in small arms ammunition boxes in accordance with LD 256008.

5.2.1.1 Data Card. Ammunition data cards, 5" x 8", in accordance with MIL-STD-1167 and printed as shown in figure I shall be properly filled out for each lot of accepted cartridges. One card shall be placed in the top of each box, two shall accompany ballistic samples, five shall be forwarded to the procuring authority and twenty shall go to the consignee.

5.3 Marking

5.3.1 Complete Cartridge. Each complete cartridge shall be marked in accordance with the applicable drawing.

5.3.2 Ammunition Box. The ammunition boxes shall be marked in accordance with Dwg. 564225.

6. NOTES

6.1 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of cartridges required.

6.2 Cartridge. A cartridge, as referred to in this specification, means a complete round of ammunition, that is, all of the component parts of the ammunition assembled into one unit necessary for firing from a gun.

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6.3 Test of Components. When the supplier for the loading and assembling of the cartridge is also the supplier of one or more components thereof the ballistic tests of the components may at the discretion of procuring agency be combined with the ballistic tests of cartridges.

6.4 General Safety Precautions. The loading, assembly and handling of the item covered by this specification and the subassemblies thereof, involve hazardous operations and therefore require suitable explosives safety precautions. Use of this specification will not be construed as to relieve the supplier or manufacturer of responsibility for the safety of his operations. The Safety provision of MIL-STD-1314 (Navy) are recommended for use by supplier in loading and assembling the items furnished under this specification. The provisions of MIL-STD-1314 are mandatory for use in Navy loading and assembly plants.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-18835C(AS) CARTRIDGES, 20MM , FOR AIRCRAFT GUNS MK 11 and MK 12

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

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